COURSE SLIDES · WSQ

ISO 14064-1:2018 Greenhouse Gas (GHG) Emissions and Removals



WSQ Course Code: TGS-2025055502

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome and Course Orientation

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Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

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About the Trainer



Your Trainer

Complete this profile before class

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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NAMED TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE

Principal Trainer and course developer

EXPERTISE

Sustainability systems, organisational GHG accounting and business improvement

DELIVERS

WSQ professional and technical programmes

FOUNDER

Tertiary Infotech / Tertiary Courses

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LEARNER INTRODUCTION

Let's Know Each Other

1

You

Name, organisation and role.

2

Your footprint

One emission source or climate issue relevant to your work.

3

Your challenge

A recurring GHG data, reporting or reduction problem.

4

Your outcome

What successful learning looks like after two days.

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HOUSEKEEPING

Ground Rules

1. **Be present**
   Return from breaks on time and complete attendance.

2. **Participate**
   Ask, challenge and contribute evidence.

3. **Respect**
   One conversation at a time; disagree professionally.

4. **Protect information**
   Use fictional or sanitised workplace data.

5. **Stay practical**
   Link every idea to a process or decision.

6. **Own the result**
   Verify your activity output before submission.

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SCHEDULE

Lesson Plan — Two Days

Day 1

- Day 1 · ISO 14064-1 inventory design, boundaries, quantification, reporting and risk controls

– 9:00-10:30 · administration, GHG science and ISO family

– 10:40-13:00 · boundaries and Activities 1-2

– 14:00-15:20 · quantification and Activity 3

– 15:30-18:00 · review, governance and Activities 4-5

Day 2

- Day 2 · Value-chain reduction, renewable energy, offsets and integrated carbon strategy

– 9:00-10:30 · hotspots, lifecycle and Activity 6

– 10:40-13:00 · supplier roadmap and renewable energy

– 14:00-15:20 · offset integrity and Activity 9

– 15:30-16:00 · Activity 10 and wrap-up

– 16:00-18:00 · WA then Case Study

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Learning Outcomes

1

LO1

Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.

2

LO2

Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.

3

LO3

Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.

4

LO4

Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

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Knowledge Alignment · K1-K5

1

K1

Evolving research on GHG dynamics and innovation in GHG accounting

2

K2

Integration of GHG accounting with sustainability and climate initiatives

3

K3

GHG reduction across product lifecycles, value chains and supply chains

4

K4

Renewable-energy technologies, economics, sustainability impacts and regulation

5

K5

Best-practice approaches for GHG accounting and reporting

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Knowledge Alignment · K6-K9

1

K6

Relevant regulations, programmes and reporting guidelines

2

K7

Frameworks for evaluating, mitigating and monitoring GHG-related risks

3

K8

Approaches and economics of carbon-offset projects

4

K9

Sustainability frameworks integrated into GHG-reduction strategy

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Ability Alignment · A1-A4

1

A1

Develop reduction strategies across product lifecycles, value chains and supply chains

2

A2

Generate the economic costs and benefits of offset projects to determine suitability

3

A3

Innovate organisational GHG accounting using emerging science and accounting practice

4

A4

Develop GHG management frameworks, plans and monitoring initiatives

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Ability Alignment · A5-A8

1

A5

Develop a carbon strategy for project selection and credit optimisation

2

A6

Influence economically viable renewable-energy strategies aligned with sustainability goals

3

A7

Develop GHG accounting and reporting guidelines integrated with sustainability initiatives

4

A8

Define assessments and mitigation strategies for GHG-related risks

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ASSESSMENT BRIEFING

Briefing for Assessment

1

Individual

No discussion; answer in your own words.

2

Open book

Use slides, Learner Guide and approved activity material only.

3

Secure

Place phones and other materials under the table; no recording.

4

Submission

Complete the provided document and upload to the LMS.

5

Competency

Every assessed K and A criterion must be Competent.

6

Rights

Re-assessment and appeal routes are available.

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FINAL ASSESSMENT

Assessment and Funding

Written Assessment

- 9 open-ended questions
- K1-K9 · 1 hour
- Open book

Case Study

- One coherent scenario
- A1-A8 · 1 hour
- Open book

Competency

- C in both instruments
- At least 75% attendance
- Funding criteria apply

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record

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SELF-CONTAINED ACTIVITY PACKS

Access the Course Activities

Option A · Clone

- Open the repository link
- Copy the HTTPS clone URL
- Open the activities folder

Option B · ZIP

- GitHub · Code
- Download ZIP
- Extract and open activities/

Open the ISO 14064-1 activity repository

Each folder

- Detailed README.md
- Working XLSX
- Printable PDF/checklist

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ACTIVITY EVIDENCE WORKFLOW

Use Evidence Before Claims

STEP 1 OF 1

1

Define the decision, document the boundary, calculate transparently, review the evidence and retain the approved output.

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https://lms-tms.tertiaryinfotech.com/

Course Portal

- Download the learner slides and Learner Guide.
- Open the complete Activities folder and working files.
- Complete and submit assessments only during the scheduled window.

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01

TOPIC 01

Foundations of Organisational GHG Accounting Systems

GHG science · ISO 14064 family · accounting principles · targets · sustainability integration

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K1 · K2 · K5 · A3 · A7 · LO1

Topic 01 · Core Model

1

Relevance

Select sources, sinks, reservoirs and methods that serve intended users and decisions.

2

Completeness

Include all significant emissions and removals within justified boundaries.

3

Consistency

Apply methods that support meaningful comparison over time and disclose changes.

4

Accuracy and transparency

Reduce bias and uncertainty, and document assumptions, evidence and limitations.

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K1 · K2 · K5 · A3 · A7 · LO1

GHG science for organisational decisions

1

Radiative forcing

GHGs absorb and re-emit infrared energy, changing the Earth's energy balance.

2

Long-lived stocks

Carbon dioxide accumulates; current emissions add to a persistent atmospheric stock.

3

Different gases

Methane, nitrous oxide and fluorinated gases differ in lifetime and warming effect.

4

CO2e

A common unit converts gas mass using a declared global-warming-potential basis.

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Decision Lens

Apply in the organisation

- Inventory all relevant gases
- State the GWP source and assessment basis
- Use science to improve priorities
- Separate physical fact from target choice

Avoid the trap

- CO2e is a comparison convention
- A lower annual total does not erase historic stock
- GWP values can change by assessment
- Avoid false precision beyond the data

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K1 · K2 · K5 · A3 · A7 · LO1

ISO 14064 family and assurance architecture

1

Part 1

Organisation-level principles and requirements for inventory design, quantification and reporting.

2

Part 2

Project-level quantification, monitoring and reporting of emission reductions or removal enhancements.

3

Part 3

Principles and requirements for verification and validation of GHG statements.

4

Related frameworks

GHG Protocol, ISO 14067 and programme rules can complement—but not silently replace—the chosen criteria.

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Decision Lens

Apply in the organisation

- Name the inventory criteria
- Define the intended assurance use
- Keep organisation and project claims distinct
- Resolve conflicting programme rules

Avoid the trap

- Certification is not automatic
- A project credit is not an organisational removal
- One framework mapping may not be exact
- Use authorised standards text for formal conformance

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K1 · K2 · K5 · A3 · A7 · LO1

ISO 14064-1 principles in practice

1

Relevance

Information and boundaries must serve intended users and decisions.

2

Completeness

Include significant sources and justify omissions transparently.

3

Consistency

Methods and boundaries support comparison through time.

4

Accuracy and transparency

Reduce bias and uncertainty and disclose assumptions, evidence and limitations.

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Decision Lens

Apply in the organisation

- Turn each principle into a review question
- Use materiality and significance deliberately
- Document changes at the time they occur
- Prefer traceable evidence over unsupported detail

Avoid the trap

- Principles interact and require judgement
- Completeness does not mean perfect data
- Consistency does not freeze weak methods
- Transparency does not excuse avoidable error

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K1 · K2 · K5 · A3 · A7 · LO1

Organisational and reporting boundaries

1

Purpose first

Intended users and decisions determine the inventory design.

2

Consolidation

Equity share, financial control or operational control changes which operations are included.

3

Reporting boundary

Direct and significant indirect emissions are organised into decision-useful categories.

4

Boundary evidence

Ownership, leases, contracts, operating authority and influence support the decision.

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Decision Lens

Apply in the organisation

- Map all entities before excluding any
- Apply one consolidation approach consistently
- Document outsourced activities
- Reassess after acquisitions or divestments

Avoid the trap

- Legal ownership and operational control can differ
- Outsourcing does not make emissions disappear
- Boundary choice can change trends
- Avoid cherry-picking low-emission operations

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K1 · K2 · K5 · A3 · A7 · LO1

Science-aligned targets and target adjustments

1

Target boundary

Specify gases, operations, categories and coverage so ambition can be compared with the inventory.

2

Base year

Choose a representative period and publish the recalculation policy before structural change occurs.

3

Pathway

Connect near-term milestones and initiatives to an explicit trajectory rather than a distant headline alone.

4

Adjustment

Recalculate for material structural or methodology changes; do not rewrite history for ordinary performance variance.

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Decision Lens

Apply in the organisation

- Link target scope to inventory scope
- Define a significance threshold
- Separate recalculation from restatement
- Retain approvals and an adjustment bridge

Avoid the trap

- A science-aligned target is not an inventory requirement
- Growth assumptions must remain visible
- Acquisitions can break comparability
- Offsets do not repair an inconsistent base year

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K1 · K2 · K5 · A3 · A7 · LO1

Quantification, base year and report integrity

1

Activity data

Measured or estimated quantities describe the emission-causing activity.

2

Emission factor

A documented coefficient translates activity into gas mass or CO2e.

3

Base year

A stable comparison point needs recalculation rules for structural and methodological change.

4

Report

Disclose boundaries, methods, exclusions, factors, uncertainty, changes and performance separately.

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Decision Lens

Apply in the organisation

- Retain a calculation trail
- Reconcile to primary records
- Recalculate under a written policy
- Report gross inventory before offsets

Avoid the trap

- Units cause frequent large errors
- Spend data can mask physical change
- Estimated data must be labelled
- Credits must not be netted into gross emissions

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Activities · Topic 1

Activity 1

- Set the GHG Inventory Purpose and Boundaries
- K1 · K2 · K5 · A3 · LO1
- 60 minutes

Activity 2

- Build a Significant GHG Source and Sink Register
- K1 · K5 · A3 · A7 · LO1
- 60 minutes

Activity 3

- Quantify a Base-Year Inventory and Reporting Guide
- K1 · K2 · K5 · A3 · A7 · LO1
- 75 minutes

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K1 · K2 · K5 · A3 · LO1 · 60 minutes

Set the GHG Inventory Purpose and Boundaries

ACTIVITY 1

Learners become the GHG inventory team for EcoForge Asia and turn an incomplete corporate structure into a justified organisational and reporting boundary.

You'll build

An inventory purpose statement, facility map, consolidation decision log and reporting-boundary diagram.

Tools: Activity workbook · EcoForge scenario pack · boundary decision checklist

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Activity 1 · Process

1. Review the fact pattern
2. Define inventory purpose
3. Map entities and operations
4. Select consolidation approach
5. Record boundary decisions

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ACTIVITY 1 · ACCEPTANCE CHECK

Set the GHG Inventory Purpose and Boundaries

✅ Test it

The diagram covers every named entity and operation; the consolidation method is applied consistently; every exclusion and unresolved item has a transparent rationale and owner.

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K1 · K5 · A3 · A7 · LO1 · 60 minutes

Build a Significant GHG Source and Sink Register

ACTIVITY 2

Learners inspect EcoForge operations and build a complete source register using ISO 14064-1 categories, with documented significance criteria.

You'll build

A source/sink register, category map, significance assessment and data-owner matrix.

Tools: Source inventory workbook · process narrative · significance criteria cards

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Activity 2 · Process

1. Map the value chain
2. Classify sources and sinks
3. Identify relevant gases
4. Define significance criteria
5. Score indirect sources

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ACTIVITY 2 · ACCEPTANCE CHECK

Build a Significant GHG Source and Sink Register

✅ Test it

The register contains all scenario sources, uses the six-category logic consistently, treats removals separately from offsets and gives evidence-based significance decisions.

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K1 · K2 · K5 · A3 · A7 · LO1 · 75 minutes

Quantify a Base-Year Inventory and Reporting Guide

ACTIVITY 3

Learners convert EcoForge activity data into CO2e, investigate anomalies and draft the accounting rules that make the inventory repeatable.

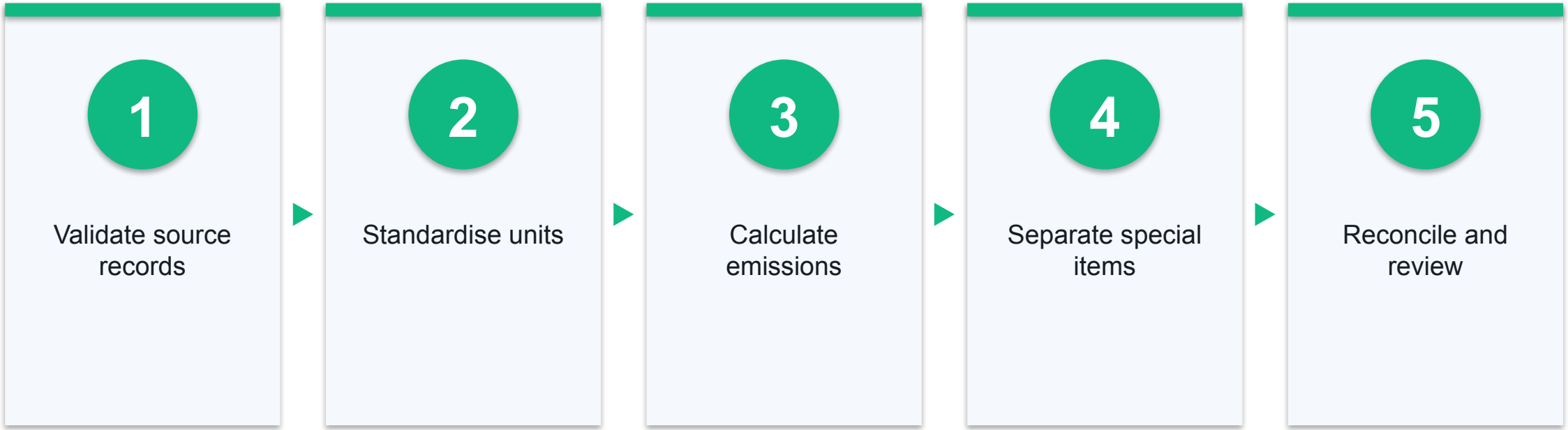
You'll build

A quantified 2025 inventory, calculation trail, uncertainty register, base-year policy and reporting-guideline extract.

Tools: GHG calculation workbook · factor register · sample invoices and meter records

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Activity 3 · Process



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ACTIVITY 3 · ACCEPTANCE CHECK

Quantify a Base-Year Inventory and Reporting Guide

✅ Test it

Every result traces to source data, units, factor and GWP basis; totals reconcile; uncertainty and recalculation triggers are explicit; no offset is netted into the inventory.

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K1 · K2 · K5 · A3 · A7 · LO1 · TOPIC CHECK

Recap · Foundations of Organisational GHG Accounting Systems

1

Explain

Select sources, sinks, reservoirs and methods that serve intended users and decisions.

2

Apply

Define intended users, reporting boundaries and consolidation choices for a credible organisational GHG inventory.

3

Demonstrate

Classify direct and indirect GHG sources, sinks and reservoirs and assess their significance for reporting.

4

Evidence

Retain the completed activity artefacts and review feedback.

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02

TOPIC 02

Regulatory-Compliant and Risk-Responsive GHG Management

Reporting requirements · inventory review · governance · climate risk · control and conformance

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K6 · K7 · A4 · A8 · LO2

Topic 02 · Core Model

1

Governance

Assign authority, competence, control ownership and escalation for inventory quality.

2

Compliance

Maintain a register of applicable laws, schemes, contracts and reporting expectations.

3

Risk

Assess data, methodological, transition, physical and reputational GHG risks.

4

Review

Use controls, reconciliation, uncertainty evaluation and management review before reporting.

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K6 · K7 · A4 · A8 · LO2

Reporting requirements and conformance

1

Applicable obligations

Laws, schemes, contracts and lender or customer terms can create different requirements.

2

Reporting criteria

The inventory must name the standard, programme rules and reporting period used.

3

Mandatory disclosures

Boundaries, categories, methods, exclusions, uncertainty and changes need controlled evidence.

4

Approval

Management accepts responsibility for the GHG statement before external use.

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Decision Lens

Apply in the organisation

- Maintain an obligation register
- Translate each requirement into evidence
- Track submission and approval dates
- Control public claim versions

Avoid the trap

- Voluntary does not mean consequence-free
- Different users may need different views
- Legal applicability is context-specific
- Do not imply verification before it occurs

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K6 · K7 · A4 · A8 · LO2

Inventory data governance and internal control

1

Ownership

Every source has an accountable owner and competent preparer.

2

Method control

Factors, GWP values, calculations and estimates are versioned and approved.

3

Evidence control

Source records are complete, retrievable, protected and retained.

4

Review control

Independent checks, reconciliations and issue closure support a reliable statement.

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Decision Lens

Apply in the organisation

- Use a source-level RACI
- Lock formulas and controlled inputs
- Log changes and approvals
- Review high-risk sources first

Avoid the trap

- Sustainability should not own all data
- A spreadsheet can hide silent changes
- Review is more than proofreading
- One approval cannot compensate for weak evidence

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K6 · K7 · A4 · A8 · LO2

Reviewing a GHG inventory report

1

Boundary test

Confirm included entities, categories and significant exclusions match documented decisions.

2

Calculation test

Reperform samples from source record through unit conversion, factor and total.

3

Disclosure test

Check claims, estimates, limitations and changes are described accurately.

4

Trend test

Challenge whether structural change, method change or data gaps explain apparent performance.

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Decision Lens

Apply in the organisation

- Trace both top-down and bottom-up
- Sample anomalies and high magnitude
- Resolve contradictions
- Retest every correction

Avoid the trap

- A balanced total can contain offsetting errors
- A factor source without vintage is weak
- Narrative claims require the same control as numbers
- Verification readiness is not verification

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K6 · K7 · A4 · A8 · LO2

GHG risk landscape

1

Inventory risk

Incomplete sources, wrong units, weak factors, estimates or uncontrolled formulas distort results.

2

Transition risk

Policy, carbon cost, technology and market expectations can affect competitiveness.

3

Physical risk

Acute and chronic hazards can disrupt operations, data and value-chain performance.

4

Claims risk

Ambiguous boundaries or unsupported neutrality and reduction statements damage trust.

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Decision Lens

Apply in the organisation

- Write cause-event-consequence statements
- Assess current-control effectiveness
- Use thresholds and early indicators
- Link material risks to management review

Avoid the trap

- Risk is not only the reported number
- High uncertainty can be decision material
- Physical and transition risks can interact
- Greenwashing risk is a governance issue

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K6 · K7 · A4 · A8 · LO2

Risk-responsive GHG management plan

1

Objectives

Connect inventory reliability and reduction outcomes to strategy and obligations.

2

Controls

Prevent, detect and correct data, method, performance and claim failures.

3

Monitoring

Use leading and lagging indicators with thresholds and escalation.

4

Learning

Review changes, incidents, findings and stakeholder challenges to improve the system.

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Decision Lens

Apply in the organisation

- Prioritise by consequence and control weakness
- Assign decision rights
- Test response scenarios
- Report residual risk transparently

Avoid the trap

- A register without action is not management
- Too many KPIs obscure warning signals
- Risk acceptance needs authority
- Plans must evolve with business change

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Activities · Topic 2

Activity 4

- Review a GHG Inventory and Resolve Quality Findings
- K5 · K6 · A4 · LO2
- 75 minutes

Activity 5

- Build a GHG Governance, Compliance and Risk Plan
- K6 · K7 · A4 · A8 · LO2
- 60 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

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K5 · K6 · A4 · LO2 · 75 minutes

Review a GHG Inventory and Resolve Quality Findings

ACTIVITY 4

Learners act as an independent internal review team and test EcoForge's draft report, calculation workbook and control evidence.

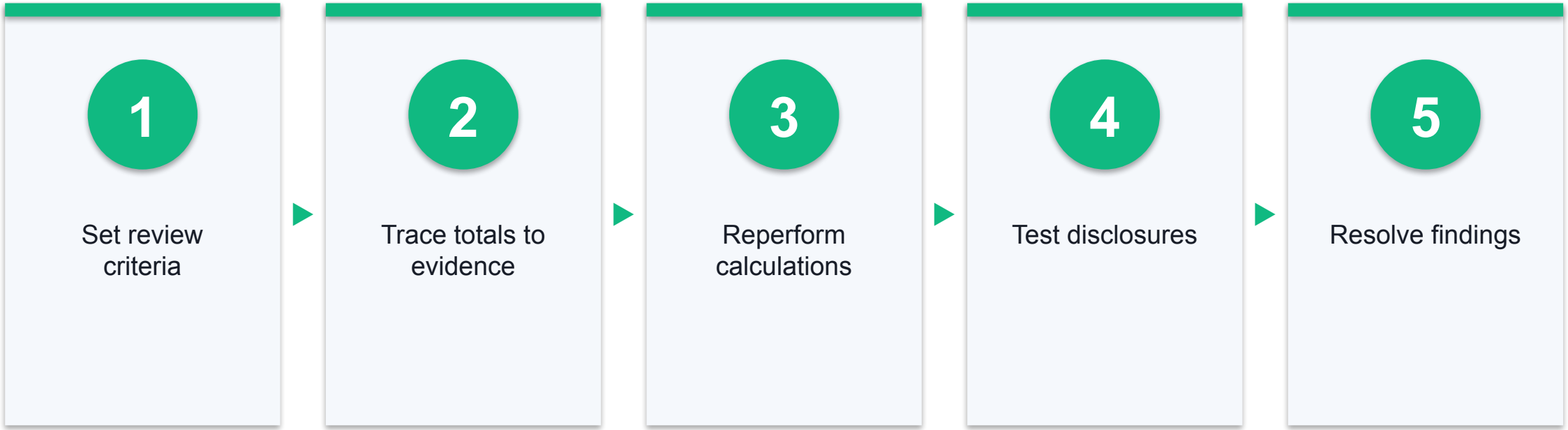
You'll build

A review plan, evidence log, finding register, corrected disclosure set and management approval brief.

Tools: Draft inventory report · evidence pack · review workbook · finding rubric

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Activity 4 · Process



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ACTIVITY 4 · ACCEPTANCE CHECK

Review a GHG Inventory and Resolve Quality Findings

✅ Test it

All seeded defects are detected; each finding is evidence-based; corrections are rechecked; the final brief distinguishes resolved issues from residual limitations.

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K6 · K7 · A4 · A8 · LO2 · 60 minutes

Build a GHG Governance, Compliance and Risk Plan

ACTIVITY 5

Learners translate EcoForge's reporting obligations and climate exposures into a practical governance and risk-management plan.

You'll build

A compliance register, RACI, GHG risk register, control plan, monitoring dashboard and escalation protocol.

Tools: Risk workbook · obligation cards · governance case pack · dashboard template

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Activity 5 · Process

1. Register obligations
2. Assign decision rights
3. Describe GHG risks
4. Assess controls
5. Plan mitigation

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ACTIVITY 5 · ACCEPTANCE CHECK

Build a GHG Governance, Compliance and Risk Plan

✅ Test it

Every applicable obligation has an owner and evidence; high risks have proportionate controls and indicators; accountability and escalation are unambiguous.

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K6 · K7 · A4 · A8 · LO2 · TOPIC CHECK

Recap · Regulatory-Compliant and Risk-Responsive GHG Management

1

Explain

Assign authority, competence, control ownership and escalation for inventory quality.

2

Apply

Evaluate inventory evidence, controls, conformance and report completeness before management approval or verification.

3

Demonstrate

Design roles, controls, compliance monitoring and mitigation actions for material GHG risks.

4

Evidence

Retain the completed activity artefacts and review feedback.

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DAY 2

Reduction Strategy → Carbon Strategy

03

TOPIC 03

Operationalising Emissions Reduction Across Value Chains

Boundaries · source data · baselines · lifecycle and supplier action · renewable energy

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K3 · K4 · A1 · A6 · LO3

Topic 03 · Core Model

1

Inventory-to-action

Use hotspot and trend evidence to direct reduction effort where it matters.

2

Hierarchy

Avoid and reduce first; substitute, recover and neutralise only where appropriate.

3

Value chain

Engage suppliers and customers using category-relevant data and decision levers.

4

Energy strategy

Compare efficiency, electrification and renewable-energy options on cost and integrity.

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K3 · K4 · A1 · A6 · LO3

From inventory to reduction hierarchy

1

Avoid

Remove unnecessary activity, demand or material before optimising it.

2

Reduce

Improve efficiency, yield, maintenance, logistics and process control.

3

Substitute

Change fuel, material, technology, product design or service model.

4

Neutralise residuals

Consider high-integrity removals or offsets only for clearly defined residual emissions.

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Decision Lens

Apply in the organisation

- Start with hotspots and influence
- Quantify expected abatement
- Check rebound and burden shift
- Separate forecast from realised result

Avoid the trap

- Largest is not always most controllable
- Low-cost action can have low impact
- Avoided emissions differ from inventory reductions
- Offsetting is not operational reduction

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K3 · K4 · A1 · A6 · LO3

Lifecycle and value-chain thinking

1

Upstream

Materials, capital goods, fuel supply, inbound logistics and supplier operations.

2

Operations

Energy, process emissions, refrigerants, waste and owned transport.

3

Downstream

Distribution, product use, maintenance, end of life and customer behaviour.

4

Interfaces

Design and procurement choices can shift emissions between stages and organisations.

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Decision Lens

Apply in the organisation

- Map physical flows before calculating
- Find decision levers at interfaces
- Use functional comparisons
- Check end-of-life assumptions

Avoid the trap

- Spend is not a lifecycle model
- Recycling claims need yield and allocation
- Efficiency can trigger rebound
- Do not double count supplier and product action

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K3 · K4 · A1 · A6 · LO3

Supplier data and engagement ladder

1

Estimate

Screen with spend, mass or industry-average factors while labelling limitations.

2

Request

Ask for activity, facility or product data using a clear specification.

3

Collaborate

Improve methods, capability and reduction plans with strategic suppliers.

4

Contract and verify

Embed criteria, evidence, improvement and assurance expectations into sourcing.

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Decision Lens

Apply in the organisation

- Segment by emissions and influence
- Standardise data requests
- Reward data improvement
- Track both coverage and reduction

Avoid the trap

- A supplier claim may use a different boundary
- Primary data can still be poor
- Questionnaires create burden
- Changing supplier may export risk rather than reduce it

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K3 · K4 · A1 · A6 · LO3

Reduction baselines, targets and monitoring

1

Baseline

State period, boundary, method, quality and structural-change treatment.

2

Target

Define absolute or intensity metric, coverage, ambition, milestones and responsibility.

3

Pathway

Model business-as-usual and initiative effects without hiding growth assumptions.

4

Monitoring

Track implementation, realised activity change, emission factors and residual gaps.

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Decision Lens

Apply in the organisation

- Use annual and interim milestones
- Reconcile projects to inventory
- Separate activity and factor effects
- Reforecast when assumptions change

Avoid the trap

- Intensity can fall while absolute emissions rise
- Targets do not prove achievement
- Forecast abatement may overlap
- Grid decarbonisation is not an internal project

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K3 · K4 · A1 · A6 · LO3

Renewable energy and economic integrity

1

Efficiency first

Reduce demand to lower cost, capacity and exposure.

2

Onsite generation

Model technical yield, self-consumption, export, maintenance and asset life.

3

Electrification

Evaluate process suitability, load, grid trajectory, resilience and capital needs.

4

Procurement

Assess contract, certificate geography, vintage, exclusivity, retirement and claim rules.

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Decision Lens

Apply in the organisation

- Compare NPV and cost per tCO2e
- Run tariff and yield sensitivities
- State inventory accounting treatment
- Stage investment with decision gates

Avoid the trap

- A certificate does not reduce physical onsite use
- Market-based claims require controls
- Cheap credits can distract from efficiency
- Technology choice must preserve safety and quality

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Activities · Topic 3

Activity 6

- Diagnose Hotspots and Set a Reduction Baseline
- K3 · A1 · LO3
- 50 minutes

Activity 7

- Design a Supplier and Product Reduction Roadmap
- K3 · K9 · A1 · LO3
- 60 minutes

Activity 8

- Develop a Renewable-Energy Business Case
- K4 · A6 · LO3
- 55 minutes

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K3 · A1 · LO3 · 50 minutes

Diagnose Hotspots and Set a Reduction Baseline

ACTIVITY 6

Learners convert EcoForge's inventory into a prioritised hotspot and abatement-opportunity view without confusing spend, emissions and influence.

You'll build

A hotspot dashboard, lifecycle map, baseline statement and opportunity shortlist.

Tools: Inventory workbook · lifecycle map · hotspot scoring tool

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Activity 6 · Process

1. Rank emission hotspots
2. Map lifecycle stages
3. Test burden shifts
4. Define the baseline
5. Prioritise reduction levers

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ACTIVITY 6 · ACCEPTANCE CHECK

Diagnose Hotspots and Set a Reduction Baseline

✅ Test it

Priorities reflect magnitude, influence and uncertainty; baseline and intensity denominator are explicit; no recommendation creates an undisclosed lifecycle burden shift.

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K3 · K9 · A1 · LO3 · 60 minutes

Design a Supplier and Product Reduction Roadmap

ACTIVITY 7

Learners build a phased roadmap that improves EcoForge's supplier data and reduces embodied, logistics and product-use emissions.

You'll build

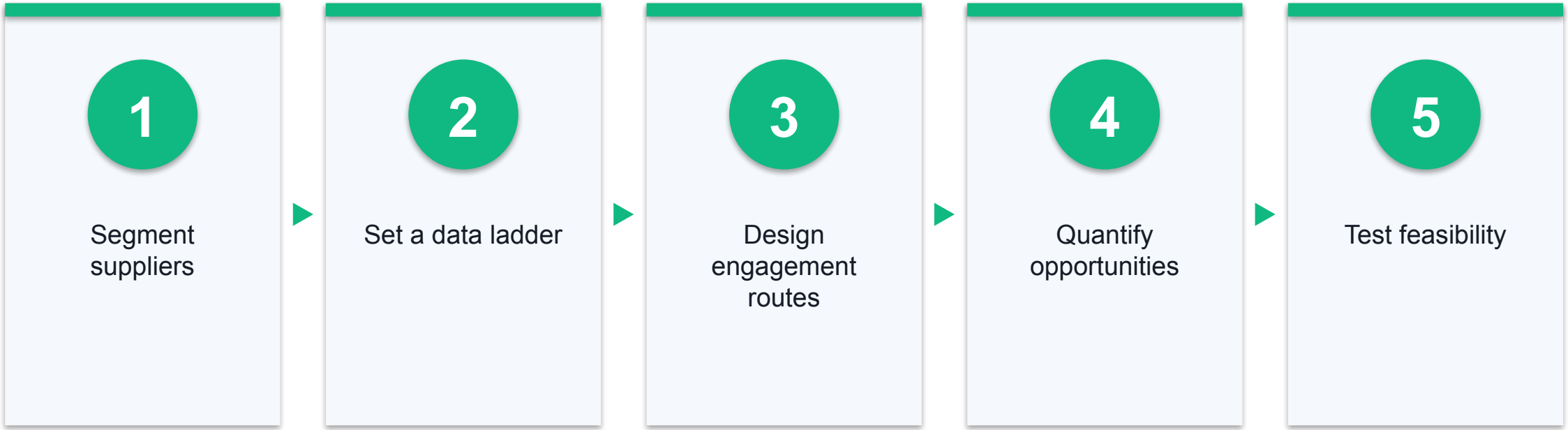
A 24-month reduction roadmap, supplier data ladder, initiative cards, KPI set and implementation governance.

Tools: Supplier portfolio workbook · initiative design cards · roadmap canvas

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K3 · K9 · A1 · LO3 · HIGH-LEVEL WORKFLOW

Activity 7 · Process



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ACTIVITY 7 · ACCEPTANCE CHECK

Design a Supplier and Product Reduction Roadmap

✅ Test it

The roadmap contains quantified, owned and time-bound actions; supplier data improves progressively; product actions address lifecycle trade-offs and claim boundaries.

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K4 · A6 · LO3 · 55 minutes

Develop a Renewable-Energy Business Case

ACTIVITY 8

Learners advise EcoForge on a portfolio of onsite solar, efficiency, electrification and renewable-electricity procurement options.

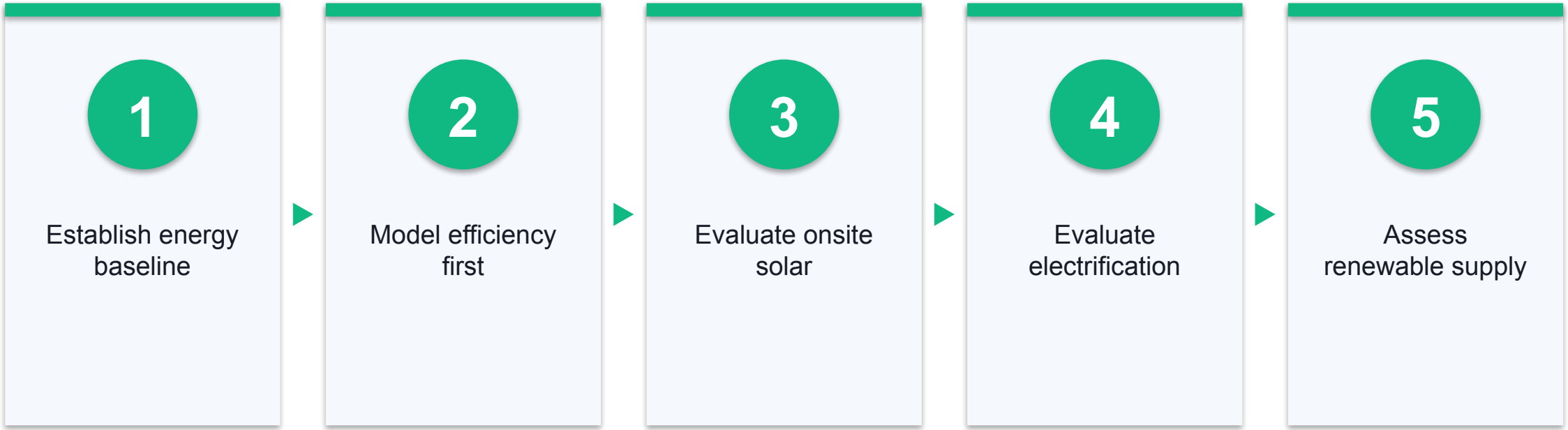
You'll build

A scenario model, marginal-abatement view, option scorecard and executive recommendation.

Tools: Energy profile workbook · tariff and option facts · financial model

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Activity 8 · Process



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ACTIVITY 8 · ACCEPTANCE CHECK

Develop a Renewable-Energy Business Case

✅ Test it

The recommendation uses consistent assumptions, separates demand and supply levers, quantifies economics and emissions, and states certificate and claim safeguards.

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Recap · Operationalising Emissions Reduction Across Value Chains

1

Explain

Use hotspot and trend evidence to direct reduction effort where it matters.

2

Apply

Use inventory, lifecycle and business data to identify hotspots, set a defensible baseline and frame reduction opportunities.

3

Demonstrate

Develop quantified value-chain actions, supplier engagement, product levers and monitoring milestones.

4

Evidence

Retain the completed activity artefacts and review feedback.

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04

TOPIC 04

Designing Viable Carbon Offset and Strategy Plans

Residual emissions · project quality · offset economics · claims · carbon-credit optimisation

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K8 · K9 · A2 · A5 · LO4

Topic 04 · Core Model

1

Residual first

Offsets address clearly defined residual emissions after credible internal reductions.

2

Quality

Evaluate additionality, permanence, leakage, quantification, verification and double counting.

3

Economics

Compare price, transaction cost, delivery risk, durability and strategic co-benefits.

4

Claims

Use precise boundaries, retirement evidence and transparent language that avoids overstatement.

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K8 · K9 · A2 · A5 · LO4

Residual emissions and the role of offsets

1

Gross inventory

Report organisational emissions and removals before external credits.

2

Reduction pathway

Show internal and value-chain actions against a declared baseline.

3

Residual emissions

Define what remains after feasible reduction within the target boundary.

4

Credit use

Disclose quantity, project, vintage, ownership, retirement and claim separately.

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Decision Lens

Apply in the organisation

- Set an offset eligibility policy
- Forecast residuals conservatively
- Procure against a clear use case
- Retire credits before claiming use

Avoid the trap

- Credits are not removals by the organisation
- Future delivery may fail
- Neutrality language can overstate scope
- Do not hide gross performance

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K8 · K9 · A2 · A5 · LO4

Carbon project integrity

1

Additionality

The claimed outcome would not occur without the project incentive under the defined test.

2

Baseline and quantification

A credible counterfactual, monitoring method and uncertainty treatment support the tonnes.

3

Permanence and leakage

Durability, reversal response and displaced emissions are addressed.

4

No double counting

Issuance, ownership, transfer, host claims and retirement prevent duplicate use.

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Decision Lens

Apply in the organisation

- Review methodology and project documents
- Check independent validation and verification
- Inspect registry status
- Assess safeguards and grievance routes

Avoid the trap

- Registry listing is not automatic quality
- Co-benefits do not repair weak carbon accounting
- Avoidance and removal have different durability
- Buffer pools do not eliminate reversal risk

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K8 · K9 · A2 · A5 · LO4

Offset economics and procurement risk

1

Unit price

Quoted cost per credit is only the starting point.

2

Total cost

Include due diligence, legal, registry, retirement, monitoring and internal governance.

3

Delivery risk

Forward projects face issuance, timing, volume, policy and counterparty uncertainty.

4

Replacement exposure

Low-quality or failed credits may require replacement at a higher future price.

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Decision Lens

Apply in the organisation

- Model risk-adjusted cost
- Use portfolio limits
- Set conditions precedent
- Retain retirement and claim evidence

Avoid the trap

- Lowest price can carry highest risk
- Co-benefits need evidence
- Vintage must match intended use
- A long contract can lock in weak criteria

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K8 · K9 · A2 · A5 · LO4

Carbon-credit portfolio optimisation

1

Purpose

Match credit type and timing to residual emissions and claim need.

2

Quality floor

Define non-negotiable methodology, verification, safeguards and registry conditions.

3

Diversification

Manage project, geography, technology, delivery and counterparty concentration.

4

Governance

Set approval, custody, retirement, accounting, disclosure and review responsibilities.

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Decision Lens

Apply in the organisation

- Use scoring plus hard exclusions
- Stage procurement over time
- Reserve for delivery failure
- Review criteria as science and policy evolve

Avoid the trap

- Diversification cannot legitimise weak credits
- Removal scarcity affects price
- Portfolio optimisation is not emissions optimisation
- Claims need boundary-specific wording

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K8 · K9 · A2 · A5 · LO4

Integrated organisational carbon strategy

1

Inventory foundation

A governed, repeatable inventory establishes what is measured and managed.

2

Target and pathway

Business growth, initiatives, dependencies and residuals are reconciled transparently.

3

Capital and risk

Investment gates balance abatement, economics, resilience, operations and data confidence.

4

Disclosure and learning

A dashboard, annual review and claim controls keep strategy credible as conditions change.

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Decision Lens

Apply in the organisation

- Connect every initiative to an inventory source
- Assign accountable owners and budgets
- Stress-test key assumptions
- Report gross, reduced and offset tonnes separately

Avoid the trap

- A strategy is not a list of projects
- Targets require governance and recalculation
- Do not combine avoided and inventory emissions
- Update claims when evidence changes

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Activities · Topic 4

Activity 9

- Evaluate Carbon Offset Projects and Economics
- K8 · K9 · A2 · LO4
- 50 minutes

Activity 10

- Integrate the Organisational Carbon Strategy
- K2 · K7 · K8 · K9 · A2 · A5 · LO4
- 30 minutes

Outputs

- Detailed procedure in LG
- Workbook and PDF pack
- Evidence-based acceptance test

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K8 · K9 · A2 · LO4 · 50 minutes

Evaluate Carbon Offset Projects and Economics

ACTIVITY 9

Learners compare three fictional project proposals and decide whether any are suitable for EcoForge's residual-emissions strategy.

You'll build

An offset due-diligence register, quality scorecard, total-cost model and procurement recommendation.

Tools: Project dossiers · quality rubric · economics workbook · claims checklist

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Activity 9 · Process

1. Define intended use
2. Review additionality
3. Assess permanence and leakage
4. Verify registry controls
5. Score safeguards and value

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ACTIVITY 9 · ACCEPTANCE CHECK

Evaluate Carbon Offset Projects and Economics

✅ Test it

The decision is evidence-based, includes quality and financial risk, requires retirement proof, and does not use credits to obscure gross organisational emissions.

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K2 · K7 · K8 · K9 · A2 · A5 · LO4 · 30 minutes

Integrate the Organisational Carbon Strategy

ACTIVITY 10

Learners complete the EcoForge capstone by presenting a 2030 carbon strategy to an executive committee under cost, growth and credibility constraints.

You'll build

A 2030 target pathway, initiative portfolio, residual-emissions forecast, offset policy, carbon-credit plan and board dashboard.

Tools: Capstone workbook · initiative portfolio · decision memo template · board role cards

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Activity 10 · Process

1. Consolidate the evidence
2. Define target architecture
3. Model the pathways
4. Sequence reduction actions
5. Govern residual emissions

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ACTIVITY 10 · ACCEPTANCE CHECK

Integrate the Organisational Carbon Strategy

✅ Test it

The strategy reconciles baseline, pathway and residuals; reduction precedes offsetting; financial and delivery risks are explicit; targets, credits and claims have accountable governance.

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Recap · Designing Viable Carbon Offset and Strategy Plans

1

Explain

Offsets address clearly defined residual emissions after credible internal reductions.

2

Apply

Assess offset quality, delivery and financial implications before procurement.

3

Demonstrate

Combine inventory governance, reduction pathways, renewable energy and high-integrity offsets into an executable strategy.

4

Evidence

Retain the completed activity artefacts and review feedback.

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WRAP-UP

From Inventory Evidence to Carbon Strategy

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COURSE OUTCOMES

What You Achieved

1

LO1

Innovate organisational GHG accounting systems using current climate science, ISO 14064-1 principles and broader sustainability initiatives.

2

LO2

Develop risk-informed GHG management plans that meet relevant reporting and regulatory requirements and support mitigation decisions.

3

LO3

Develop operational emissions-reduction strategies across product lifecycles, value chains, supply chains and renewable-energy options.

4

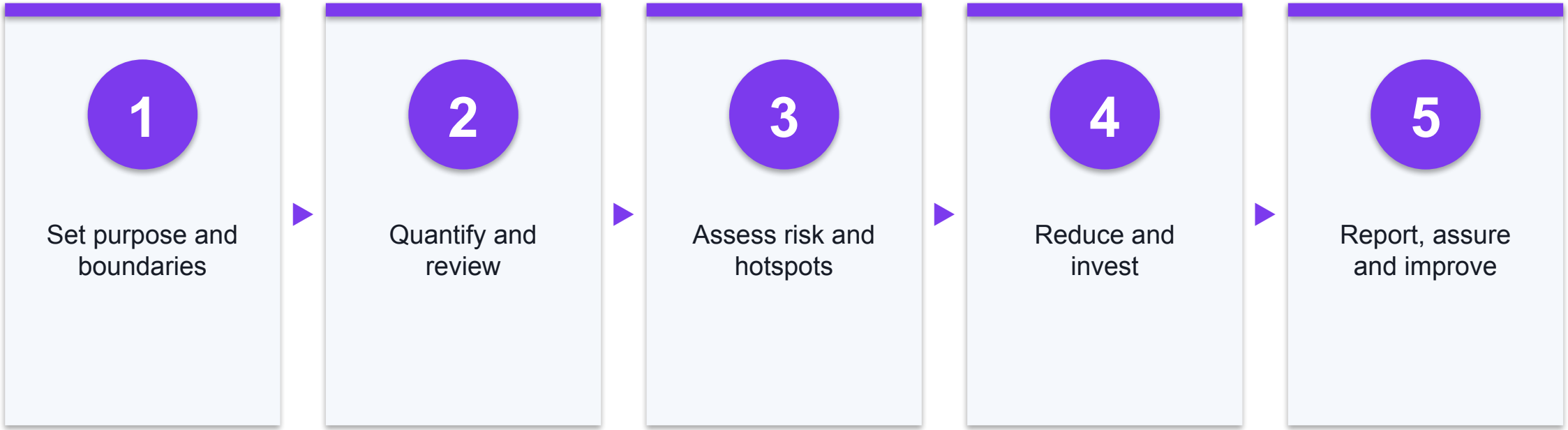
LO4

Generate economically viable carbon-offset and carbon-credit strategies aligned with organisational targets and residual-emissions needs.

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END-TO-END PRACTICE

The Complete GHG Management Cycle



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https://lms-tms.tertiaryinfotech.com/

Courseware and Assessment on the LMS

1

Download

Get the learner slides PDF and Learner Guide.

2

Activities

Open the complete Activities folder and working files.

3

Assessment

Complete the WA and Case Study during the scheduled window.

4

Submit

Upload completed answers and confirm successful submission.

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FINAL REMINDER

Assessment

WA

- 9 questions
- K1-K9
- 1 hour · open book

Case Study

- 4 questions
- A1-A8
- 1 hour · open book

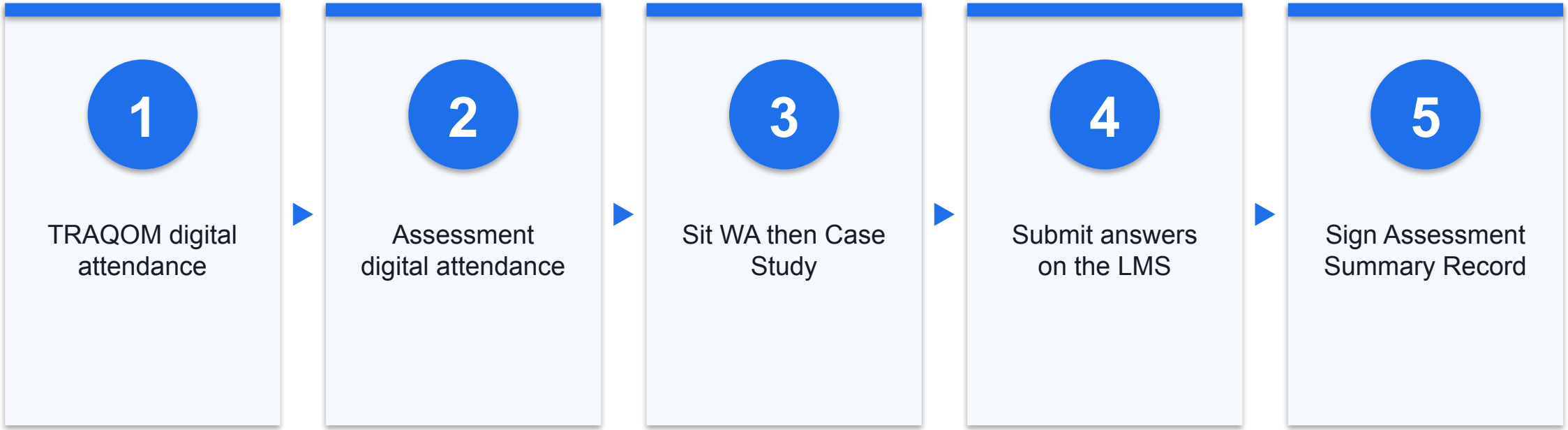
Complete

- Assessment attendance
- Submit on LMS
- Sign summary record

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ON ASSESSMENT DAY

Assessment Flow



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Digital Attendance (Mandatory)

1

AM, PM and assessment

Complete every required SSG digital-attendance session.

2

Scan

Use the SSG QR code shown by the trainer or administrator.

3

Confirm

Wait for the successful submission acknowledgement.

4

Eligibility

At least 75% attendance is required for assessment and funding.

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THANK YOU

Account for what matters. Reduce what you can. Disclose what remains.

Use transparent evidence to turn an organisational inventory into credible climate decisions.

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